

Chapter 13: Sampling Distributions

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Part III: Foundations of Inference

Why This Chapter Matters

A random sample gives information about a population, but a different random sample usually gives a different answer.

Chapter 13 explains how that natural variation behaves. This idea is the bridge from descriptive statistics to statistical inference.

By the end of the chapter, the main questions should be clear.

- What is the difference between a parameter and a statistic
- What is a sampling distribution
- Why do larger samples give more stable results
- When can a Normal model be used for a sample mean or a sample proportion

Parameters and Statistics

A **parameter** is a numerical characteristic of a population.

A **statistic** is a numerical characteristic calculated from sample data.

Measure	Sample statistic	Population parameter
Mean	$\bar{x}$	μ
Proportion	$\hat{p}$	p

The symbols matter because they tell us whether a number comes from the sample or from the full population.

Population, Sample and Sampling Distribution

Three related ideas appear throughout this chapter.

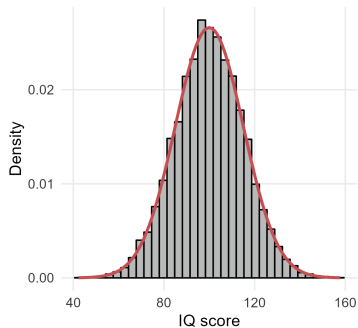
- The **population distribution** describes values for individuals in the population
- The **sample distribution** describes the observations in one particular sample
- The **sampling distribution** describes the values of a statistic across many possible samples

The most common confusion in this chapter is mixing up the distribution of individual observations with the distribution of a statistic such as $\bar{x}$.

Population, Sample and Sampling Distribution

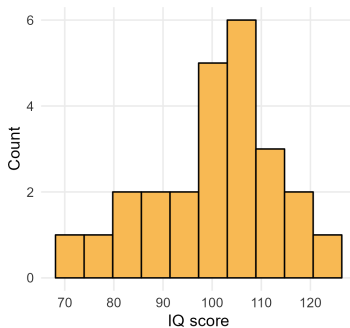
Population Distribution

Individual IQ scores



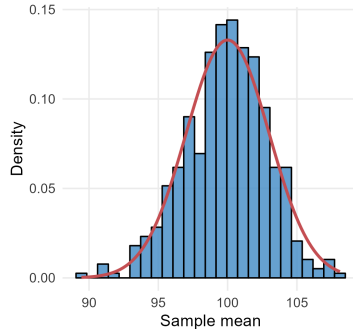
One Sample Distribution

The 25 observed IQ scores in one sample



Sampling Distribution

Five hundred sample means from samples of size 25



Example 13.1: A Population Parameter

Question

A national registration system contains birth records for every birth in the population of interest during one year. The records show that 8% of births were low birth weight and that the mean age at first birth was 26.3 years. Are these values parameters or statistics

Example 13.1: A Population Parameter

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Explanation

These values were calculated from the entire population defined for the study rather than from a sample.

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Explanation

These values were calculated from the entire population defined for the study rather than from a sample.

Solution

Both values are **parameters**.

$$p = 0.08$$

$$\mu = 26.3$$

Example 13.2: A Sample Statistic

Question

A polling organization interviews a random sample of 1023 adults. In the sample, 19% report smoking cigarettes during the previous week. Among the sampled smokers, the mean number of cigarettes per day is 12. Are these values parameters or statistics

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Explanation

The values were computed from a sample rather than from every adult in the target population.

Solution

Both values are **statistics**.

$$\hat{p} = 0.19$$

$$\bar{x} = 12$$

Repeated Sampling and the Sampling Distribution

A sample mean is not fixed before the sample is selected. It depends on which individuals happen to be chosen.

A **sampling distribution** answers a repeated-sampling question.

Sampling Distribution

The sampling distribution of a statistic is the probability distribution of that statistic over all possible random samples of the same size drawn under the same sampling process.

A computer simulation can approximate this idea by drawing many samples and calculating the statistic each time.

Examples 13.3 and 13.4: Simulating Sample Means

Question

Individual IQ scores are modelled by a Normal distribution with mean $\mu = 100$ and standard deviation $\sigma = 15$. What happens if 500 random samples of size $n = 25$ are selected and the sample mean is calculated for each sample

Examples 13.3 and 13.4: Simulating Sample Means

Question

Individual IQ scores are modelled by a Normal distribution with mean $\mu = 100$ and standard deviation $\sigma = 15$. What happens if 500 random samples of size $n = 25$ are selected and the sample mean is calculated for each sample

Explanation

Each sample contributes one number to the sampling distribution, namely its sample mean. Because the population is Normal, the sampling distribution of $\bar{x}$ is also Normal.

$$\mu_{\bar{x}} = 100$$

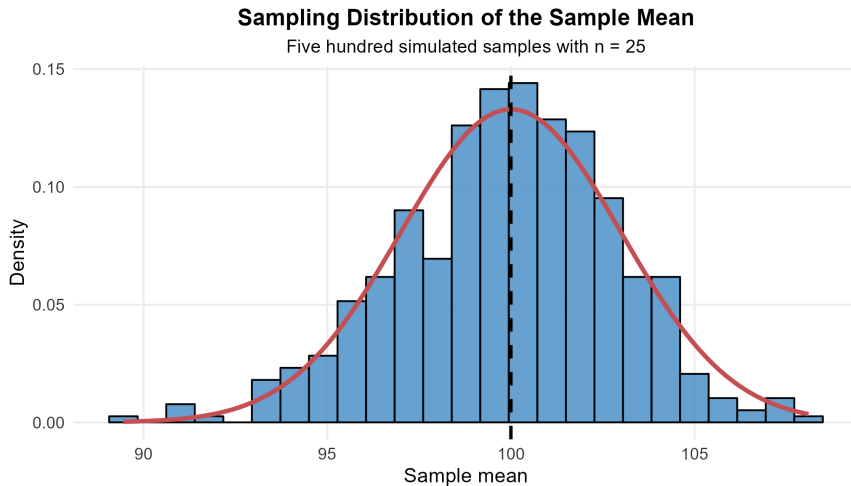
$$\sigma_{\bar{x}} = \frac{15}{\sqrt{25}} = 3$$

Examples 13.3 and 13.4: Simulating Sample Means

Solution

The histogram of the 500 simulated means is centred near 100 and has much less spread than the distribution of individual IQ scores.

Examples 13.3 and 13.4: Simulated Sampling Distribution



The Sampling Distribution of $\bar{x}$

If a random sample of size n is drawn from a population with mean μ and standard deviation σ , then the sampling distribution of $\bar{x}$ has two key properties.

Centre

$$\mu_{\bar{x}} = \mu$$

Spread

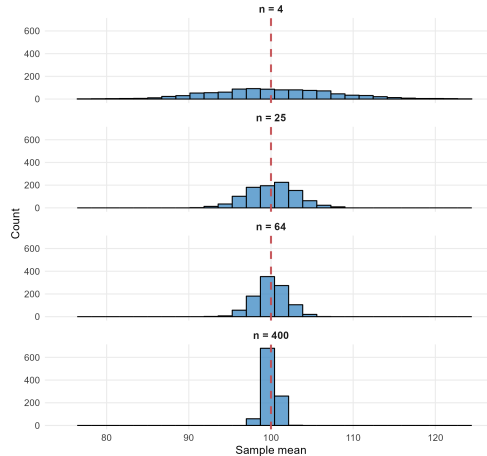
$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

The sample mean is therefore an **unbiased estimator** of the population mean. Larger samples make the sampling distribution narrower.

The Effect of Sample Size

How Sample Size Affects the Spread of the Sample Mean

Larger samples produce a narrower sampling distribution



Why the Square Root Relationship Matters

The standard deviation of the sample mean is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

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If the sample size is multiplied by 4, then

$$\frac{\sigma}{\sqrt{4n}} = \frac{1}{2} \frac{\sigma}{\sqrt{n}}$$

Why the Square Root Relationship Matters

The standard deviation of the sample mean is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

If the sample size is multiplied by 4, then

$$\frac{\sigma}{\sqrt{4n}} = \frac{1}{2} \frac{\sigma}{\sqrt{n}}$$

If the sample size is multiplied by 100, then

$$\frac{\sigma}{\sqrt{100n}} = \frac{1}{10} \frac{\sigma}{\sqrt{n}}$$

Why the Square Root Relationship Matters

The standard deviation of the sample mean is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

If the sample size is multiplied by 4, then

$$\frac{\sigma}{\sqrt{4n}} = \frac{1}{2} \frac{\sigma}{\sqrt{n}}$$

If the sample size is multiplied by 100, then

$$\frac{\sigma}{\sqrt{100n}} = \frac{1}{10} \frac{\sigma}{\sqrt{n}}$$

A larger sample improves precision, but the improvement is slower than many students expect.

When Is the Sampling Distribution Normal

The shape of the sampling distribution of $\bar{x}$ depends on the population and the sample size.

Two important cases appear in first-year statistics.

- If the population is Normal, then the sampling distribution of $\bar{x}$ is exactly Normal for any sample size
- If the population is not Normal, the Central Limit Theorem tells us that the sampling distribution of $\bar{x}$ becomes approximately Normal when the sample size is sufficiently large

Example 13.5: Population Values and Sample Means

Question

Individual IQ scores are modelled by a Normal distribution with $\mu = 100$ and $\sigma = 15$. Random samples of size $n = 25$ are selected. How does the population distribution compare with the sampling distribution of $\bar{x}$

Example 13.5: Population Values and Sample Means

Question

Individual IQ scores are modelled by a Normal distribution with $\mu = 100$ and $\sigma = 15$. Random samples of size $n = 25$ are selected. How does the population distribution compare with the sampling distribution of $\bar{x}$

Explanation

Both distributions are centred at 100. The difference is the spread.

$$\sigma = 15$$

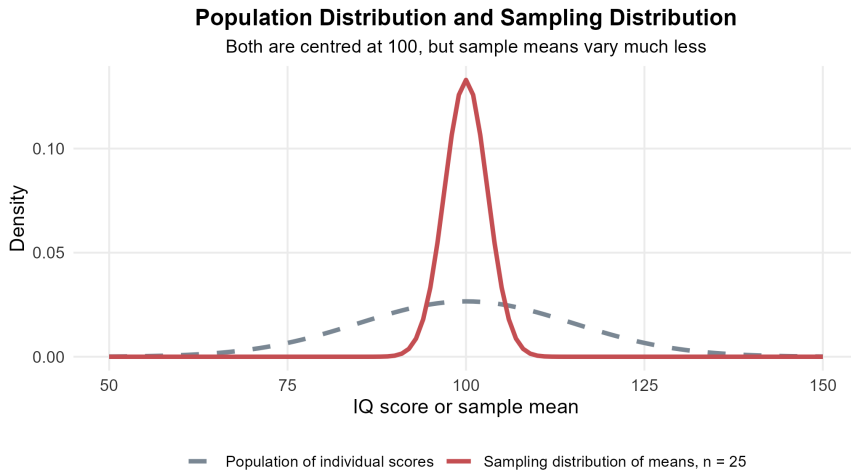
$$\sigma_{\bar{x}} = \frac{15}{\sqrt{25}} = 3$$

Example 13.5: Population Values and Sample Means

Solution

The sampling distribution of sample means is much narrower than the population distribution of individual scores.

Example 13.5: Population and Sampling Distributions



The Central Limit Theorem

Central Limit Theorem

For independent observations from a population with finite mean μ and finite standard deviation σ , the sampling distribution of $\bar{x}$ becomes approximately Normal as the sample size becomes large.

The theorem does **not** say that the raw observations in one large sample become Normal.

The theorem does say that the distribution of **sample means across repeated samples** becomes more nearly Normal as n increases.

Example 13.6: The Central Limit Theorem in Action

Question

A population is strongly right-skewed. How does the shape of the sampling distribution of $\bar{x}$ change as the sample size increases

Example 13.6: The Central Limit Theorem in Action

Question

A population is strongly right-skewed. How does the shape of the sampling distribution of $\bar{x}$ change as the sample size increases

Explanation

A sample mean averages several observations. As more observations are included, the distribution of the mean becomes less skewed.

Example 13.6: The Central Limit Theorem in Action

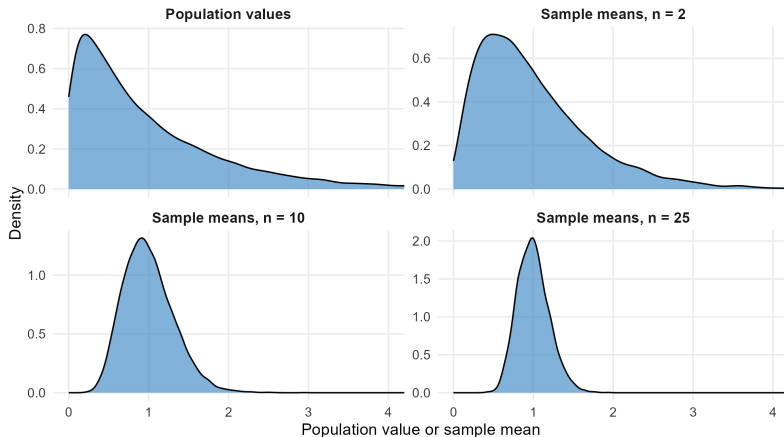
Solution

The population remains right-skewed. The sampling distribution of the sample mean becomes more symmetric and more nearly Normal as the sample size grows.

Example 13.6: The Central Limit Theorem in Action

The Central Limit Theorem with a Right-Skewed Population

The distribution of sample means becomes more nearly Normal as n increases



Assessing Whether a Normal Approximation Is Reasonable

A practical first-year approach is to consider three things together.

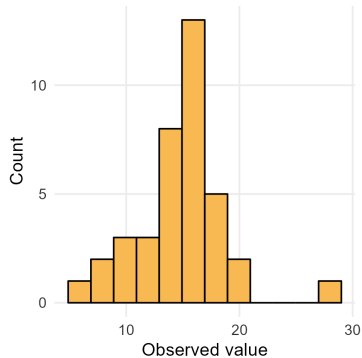
- The shape of the population or sample histogram
- The sample size
- Whether severe outliers or strong skewness are present

A mild departure from symmetry often requires less averaging than a severe departure from symmetry.

Illustrative Shapes for CLT Discussion

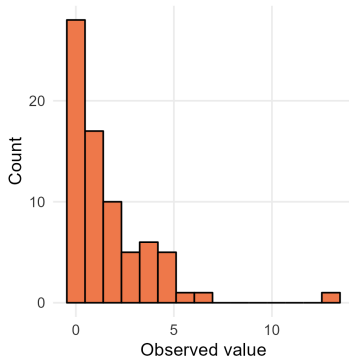
A. Mostly Symmetric

Illustrative $n = 38$



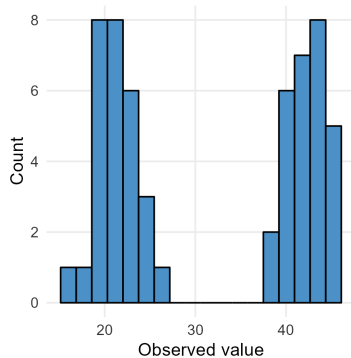
B. Strongly Right-Skewed

Illustrative $n = 74$



C. Bimodal

Illustrative $n = 56$



The Sample Proportion

For a categorical variable with two outcomes, the statistic of interest is often the sample proportion.

If X is the number of successes in a sample of size n , then

$$\hat{p} = \frac{X}{n}$$

The word success is only a label for the outcome being counted.

The Sampling Distribution of $\hat{p}$

The sampling distribution of $\hat{p}$ has properties similar to the sampling distribution of $\bar{x}$.

Centre

$$\mu_{\hat{p}} = p$$

Spread

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

A Normal approximation is typically used when both expected counts are at least 10.

$$np \geq 10$$

$$n(1-p) \geq 10$$

Example 13.8: Contact Lens Practices

Question

A random sample of 281 contact-lens users contains 139 people who do not consistently wash their hands before handling their lenses. What is the sample proportion

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Explanation

The count of successes is $X = 139$ and the sample size is $n = 281$. The sample proportion is the count divided by the sample size.

Example 13.8: Contact Lens Practices

Question

A random sample of 281 contact-lens users contains 139 people who do not consistently wash their hands before handling their lenses. What is the sample proportion

Explanation

The count of successes is $X = 139$ and the sample size is $n = 281$. The sample proportion is the count divided by the sample size.

Solution

$$\hat{p} = \frac{139}{281}$$
$$\hat{p} \approx 0.495$$

The sample proportion is about 49.5%.

Examples 13.9 and 13.10: Blood Type O

Question

Suppose the population proportion with blood type O is $p = 0.45$. A random sample of $n = 1000$ records is selected. What is the probability that the sample proportion is 0.42 or lower

Examples 13.9 and 13.10: Blood Type O

Question

Suppose the population proportion with blood type O is $p = 0.45$. A random sample of $n = 1000$ records is selected. What is the probability that the sample proportion is 0.42 or lower

Step 1: Identify the Sampling Distribution

The centre is

$$\mu_{\hat{p}} = 0.45$$

The standard deviation is

$$\sigma_{\hat{p}} = \sqrt{\frac{0.45(0.55)}{1000}} \approx 0.01573$$

The success-failure condition is satisfied because $np = 450$ and $n(1 - p) = 550$.

Examples 13.9 and 13.10: Blood Type O

Step 2: Convert the Question to a Probability

The probability of interest is

$$P(\hat{p} \leq 0.42)$$

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Step 3: Standardize

$$z = \frac{0.42 - 0.45}{0.01573} \approx -1.91$$

Examples 13.9 and 13.10: Blood Type O

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$$z = \frac{0.42 - 0.45}{0.01573} \approx -1.91$$

Solution

Using the Normal model gives

$$P(\hat{p} \leq 0.42) \approx 0.0283$$

This is about 2.83%.

Examples 13.9 and 13.10: R Code

The same probability can be calculated in R with `pnorm()`.

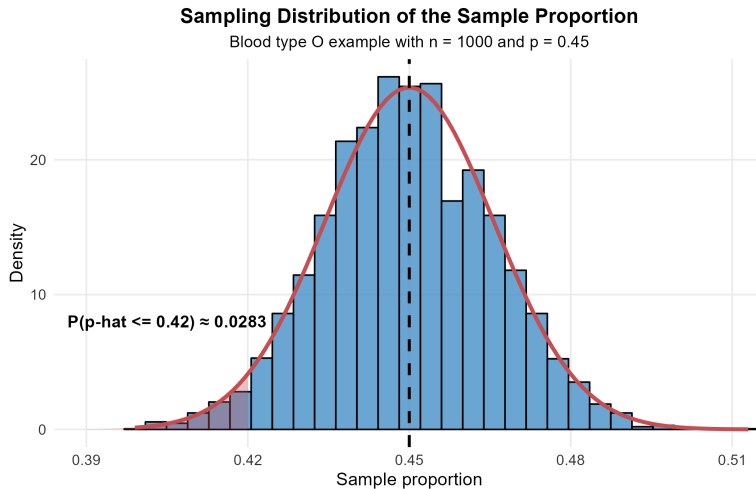
```
# Population proportion and sample size
p_type_o <- 0.45
n_type_o <- 1000

# Standard deviation of the sampling distribution of p-hat
sd_type_o <- sqrt(p_type_o * (1 - p_type_o) / n_type_o)

# Lower-tail probability  $P(\hat{p} \leq 0.42)$ 
prob_type_o <- pnorm(
  q = 0.42,
  mean = p_type_o,
  sd = sd_type_o
)

print(round(prob_type_o, 4))
```

Sampling Distribution of $\hat{p}$



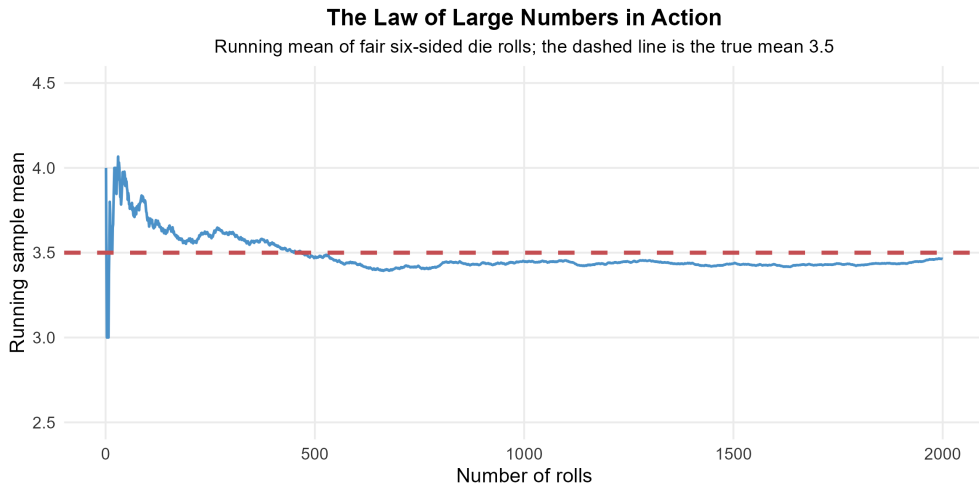
The Law of Large Numbers

Law of Large Numbers

For independent observations with a finite expected value, the sample mean converges to the population mean as the sample size increases. For independent Bernoulli observations, the sample proportion converges to the population probability.

This is a long-run idea. It does not say that every new observation brings the statistic closer to the parameter. It says that the overall tendency is toward the true population value.

The Law of Large Numbers in Action



Why the Law of Large Numbers Matters

The Law of Large Numbers explains why averages become more stable in large collections of observations.

Examples include

- Long-run average claim costs in insurance portfolios
- Long-run average outcomes in repeated physical or biological measurements
- Stabilization of a running proportion or a running mean in simulation studies

In the life sciences, sample sizes are often limited by cost, time or ethics, which is why sampling distributions are so important.

Practice Example: Sampling Distribution of $\bar{x}$

Question

A genetic strain of laboratory mice has population mean gestation period $\mu = 20$ days and population standard deviation $\sigma = 1.5$ days. A random sample contains $n = 15$ mice. What are the mean and standard deviation of the sampling distribution of $\bar{x}$

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Explanation

The centre of the sampling distribution equals the population mean. The spread is the population standard deviation divided by the square root of the sample size.

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Explanation

The centre of the sampling distribution equals the population mean. The spread is the population standard deviation divided by the square root of the sample size.

Solution

$$\begin{aligned}\mu_{\bar{x}} &= 20 \\ \sigma_{\bar{x}} &= \frac{1.5}{\sqrt{15}} \approx 0.387\end{aligned}$$

Practice Example: Normal Population Probability

Question

Blood cholesterol levels in healthy adult Golden Retrievers are modelled as Normal with mean $\mu = 200$ mg/dL and standard deviation $\sigma = 15$ mg/dL. A random sample contains $n = 9$ dogs. What is the probability that the sample mean exceeds 210 mg/dL

Practice Example: Normal Population Probability

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Blood cholesterol levels in healthy adult Golden Retrievers are modelled as Normal with mean $\mu = 200$ mg/dL and standard deviation $\sigma = 15$ mg/dL. A random sample contains $n = 9$ dogs. What is the probability that the sample mean exceeds 210 mg/dL

Step 1: Describe the Sampling Distribution

Because the population is Normal, the sampling distribution of $\bar{x}$ is Normal.

$$\mu_{\bar{x}} = 200$$

$$\sigma_{\bar{x}} = \frac{15}{\sqrt{9}} = 5$$

Practice Example: Normal Population Probability

Step 2: State the Probability

$$P(\bar{x} > 210)$$

Practice Example: Normal Population Probability

Step 2: State the Probability

$$P(\bar{x} > 210)$$

Solution

Using the Normal model gives

$$P(\bar{x} > 210) \approx 0.0228$$

This is about 2.28%.

Practice Example: Normal Population Probability R Code

```
# Sampling standard deviation for x-bar
sampling_sd <- 15 / sqrt(9)

# Upper-tail probability P(x-bar > 210)
prob_cholesterol <- pnorm(
  q = 210,
  mean = 200,
  sd = sampling_sd,
  lower.tail = FALSE
)

print(round(prob_cholesterol, 4))
```

Practice Example: CLT Probability

Question

A population of viral-load measurements is strongly right-skewed with mean $\mu = 1000$ particles/mL and standard deviation $\sigma = 300$ particles/mL. A random sample contains $n = 50$ patients. What is the approximate probability that the sample mean is below 900 particles/mL

Practice Example: CLT Probability

Question

A population of viral-load measurements is strongly right-skewed with mean $\mu = 1000$ particles/mL and standard deviation $\sigma = 300$ particles/mL. A random sample contains $n = 50$ patients. What is the approximate probability that the sample mean is below 900 particles/mL?

Explanation

The population is not Normal, but the sample size is reasonably large. The Central Limit Theorem supports an approximate Normal model for $\bar{x}$.

$$\mu_{\bar{x}} = 1000$$

$$\sigma_{\bar{x}} = \frac{300}{\sqrt{50}} \approx 42.43$$

Practice Example: CLT Probability

Solution

The probability of interest is

$$P(\bar{x} < 900)$$

Using the approximate Normal model gives

$$P(\bar{x} < 900) \approx 0.0092$$

This is about 0.92%.

Practice Example: CLT Probability R Code

```
# Approximate sampling standard deviation from the CLT
sampling_sd_viral <- 300 / sqrt(50)

# Lower-tail probability  $P(\bar{x} < 900)$ 
prob_viral <- pnorm(
  q = 900,
  mean = 1000,
  sd = sampling_sd_viral
)

print(round(prob_viral, 4))
```

Practice Example: Sample Proportion Probability

Question

An antibiotic has cure probability $p = 0.80$. A random sample contains $n = 200$ infected patients. What is the approximate probability that fewer than 75% of the patients are cured

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An antibiotic has cure probability $p = 0.80$. A random sample contains $n = 200$ infected patients. What is the approximate probability that fewer than 75% of the patients are cured

Step 1: Check Conditions and Describe the Sampling Distribution

The Normal approximation is appropriate because

$$np = 160$$

$$n(1 - p) = 40$$

The centre and spread are

$$\mu_{\hat{p}} = 0.80$$

$$\sigma_{\hat{p}} = \sqrt{\frac{0.80(0.20)}{200}} \approx 0.0283$$

Practice Example: Sample Proportion Probability

Step 2: State the Probability

$$P(\hat{p} < 0.75)$$

Practice Example: Sample Proportion Probability

Step 2: State the Probability

$$P(\hat{p} < 0.75)$$

Solution

Using the Normal model gives

$$P(\hat{p} < 0.75) \approx 0.0385$$

This is about 3.85%.

Practice Example: Sample Proportion Probability R Code

```
# Model parameters for the sample proportion
p_cure <- 0.80
n_cure <- 200
sd_cure <- sqrt(p_cure * (1 - p_cure) / n_cure)

# Lower-tail probability  $P(\hat{p} < 0.75)$ 
prob_cure <- pnorm(
  q = 0.75,
  mean = p_cure,
  sd = sd_cure
)

print(round(prob_cure, 4))
```

Preview of Statistical Inference

Question

A baseline model gives mean recovery time $\mu = 14$ days with standard deviation $\sigma = 3$ days. A random sample of $n = 100$ patients receiving a new treatment has mean recovery time $\bar{x} = 13.1$ days. If the baseline model were still true, how unusual would a sample mean of 13.1 days or lower be

Preview of Statistical Inference

Question

A baseline model gives mean recovery time $\mu = 14$ days with standard deviation $\sigma = 3$ days. A random sample of $n = 100$ patients receiving a new treatment has mean recovery time $\bar{x} = 13.1$ days. If the baseline model were still true, how unusual would a sample mean of 13.1 days or lower be

Explanation

Under the baseline model, the sampling distribution of $\bar{x}$ has

$$\mu_{\bar{x}} = 14$$

$$\sigma_{\bar{x}} = \frac{3}{\sqrt{100}} = 0.30$$

Preview of Statistical Inference

Solution

The probability of interest is

$$P(\bar{x} \leq 13.1)$$

Using the Normal model gives

$$P(\bar{x} \leq 13.1) \approx 0.0013$$

Solution

The probability of interest is

$$P(\bar{x} \leq 13.1)$$

Using the Normal model gives

$$P(\bar{x} \leq 13.1) \approx 0.0013$$

A small probability means that the observed sample mean would be unusual if the baseline model were true. This repeated-sampling logic is the basis of formal hypothesis testing in later chapters.

Preview of Statistical Inference R Code

```
# Baseline model for the sampling distribution of x-bar
baseline_mean <- 14
population_sd <- 3
sample_size <- 100
sampling_sd_recovery <- population_sd / sqrt(sample_size)

# Lower-tail probability  $P(\bar{x} \leq 13.1)$ 
prob_recovery <- pnorm(
  q = 13.1,
  mean = baseline_mean,
  sd = sampling_sd_recovery
)

print(round(prob_recovery, 4))
```

R Functions Used in This Lecture

Several R functions appear in the notes and examples.

- `pnorm()` for Normal probabilities
- `sqrt()` for square roots in standard-deviation formulas
- `round()` for rounded numerical output
- `rnorm()` for simulated Normal data in the notes
- `rbinom()` for simulated binomial counts in the notes

The lecture figures in the notes also use `ggplot()` and related plotting layers to visualize sampling distributions.

Chapter Summary

- A parameter describes a population and a statistic is calculated from a sample
- A sampling distribution describes how a statistic behaves across repeated random samples
- For a sample mean, the centre is μ and the spread is $\sigma/\sqrt{n}$
- For a sample proportion, the centre is p and the spread is $\sqrt{p(1-p)/n}$
- The Central Limit Theorem explains why sample means become approximately Normal for large samples
- The Law of Large Numbers explains why sample statistics stabilize near the true population value in the long run